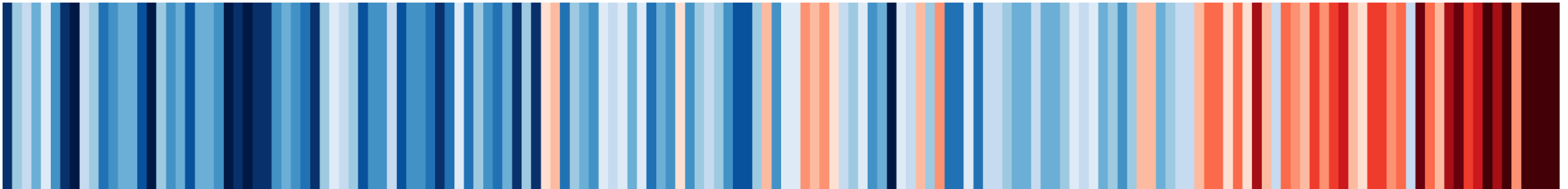


Psychology of Climate Change Communication

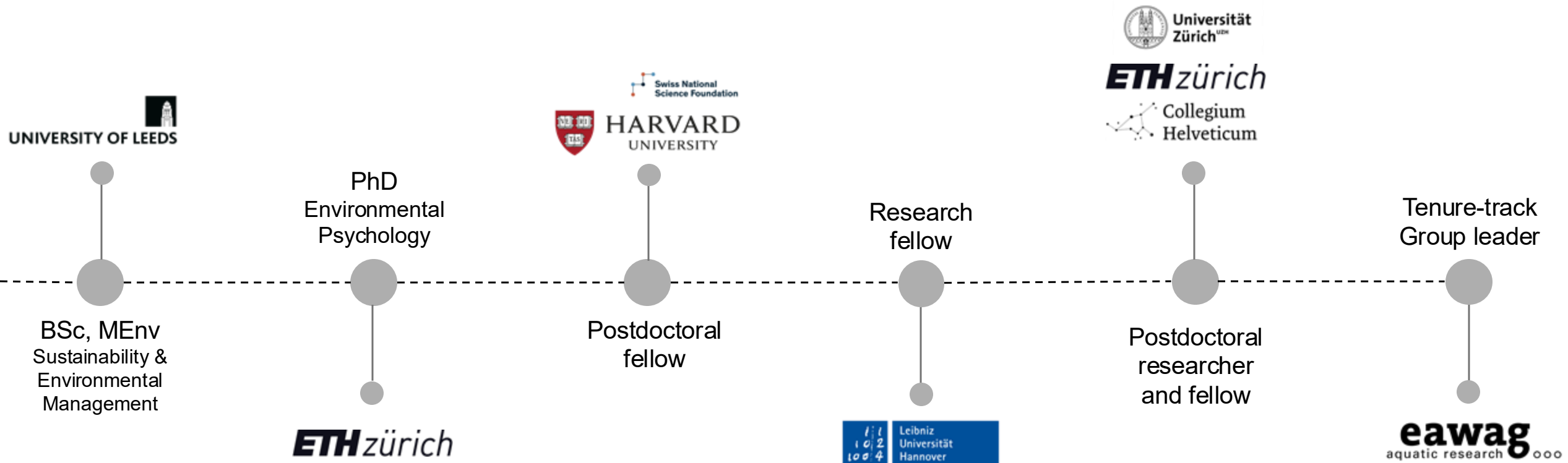
Week 1: Introduction

Dr. Viktoria Cologna and Dr. Jan Pfänder

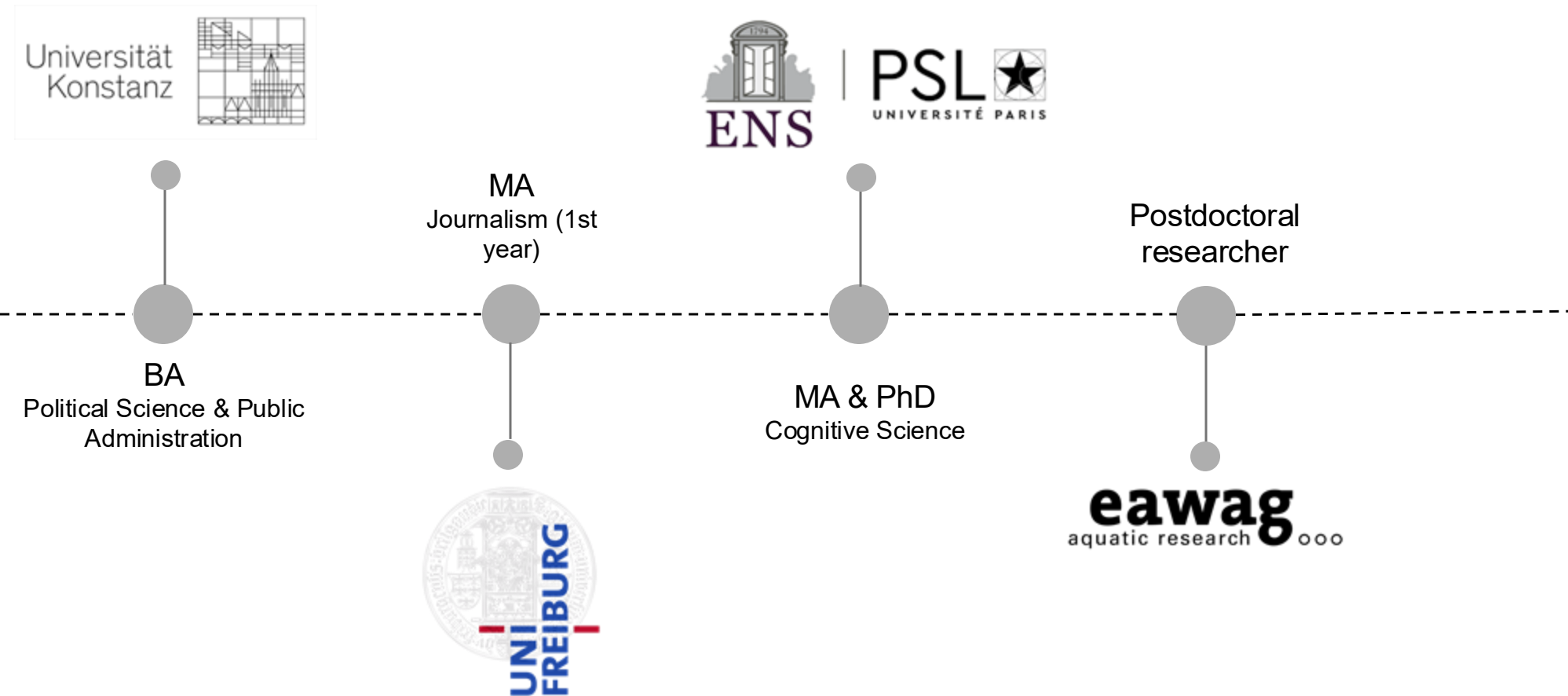


Who we are

Dr. Viktoria Cologna (she/her)



Dr. Jan Pfänder (he/him)

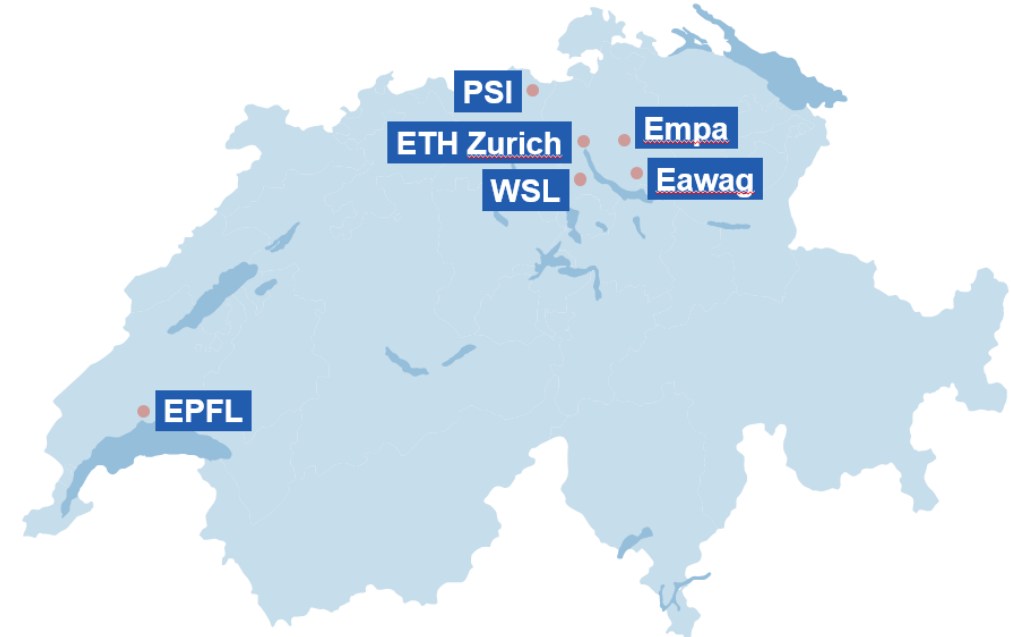


Where we work: Eawag

Department of Environmental Social Sciences



The ETH Domain



Climate and societal change group

(Future) Team members



Dr. Jan Pfänder
Postdoc
Focus: Trust in science



Dr. Charlotte Kukowski
Postdoc
Focus: Env. psychology



Dr. Tobia Spampatti
Postdoc
Focus: Misinformation



Zahra Rahmani
Postdoc
Focus: Climate change communication

Our broad research interests and expertise

- Environmental psychology
- Behavioral science
- Trust in science
- Misinformation
- Climate change communication

Research questions we are currently pursuing

- What messages are effective at increasing trust in climate scientists?
- How does trust in scientists relate to engagement on climate change?
- How do people around the world feel about climate change?
- How has news media reporting on climate change evolved?
- Are people able to distinguish true from false news?

Our teaching philosophy

- Inclusion
- Stimulating interest in the subject matter
- Critical thinking

Meet with us

- Questions on lecture
- Feedback
- Bachelor and master thesis

Who are you?

- Name
- Pronouns if you wish
- Study program and level
- Why you enrolled in this course
- Last/favorite TV show you watched

Course information

When and where: Tuesday, 16:15-18:00, CHN G42

Break: 17:00-17:15

Credits: 3 ECTS

Moodle: Syllabus, reading materials, slides

Assessment: Pass or fail

No recording will be available. In-person presence expected.

Course information

Important: Participation in this course requires a course-internal commitment at the latest by Friday, September 25 (23h59).

After this date, groups will be fixed, and students are expected to remain in the course. Students who do not intend to participate must delete the course from myStudies by the commitment deadline.

Course overview

#	Date	Topic	Lecturer
1	15.09.2026	Introduction and key concepts in environmental psychology	Viktoria Cologne & Jan Pfänder
2	22.09.2026	Public opinion on climate change and the deficit model	Viktoria Cologne
3	29.09.2026	A question of trust	Viktoria Cologne
4	06.10.2026	Mental models and framing	Jan Pfänder
5	13.10.2026	Climate change emotions	Viktoria Cologne
6	20.10.2026	Guest speaker: Dr. Cornelia Schwierz	Invited speaker
7	27.10.2026	Guest speaker: Dr. Reto Knutti	Invited speaker

Course overview

#	Date	Topic	Lecturer
8	03.11.2026	Climate change in the media	Viktoria Cologne
9	10.11.2026	Climate change misinformation	Jan Pfänder
10	17.11.2026	Guest speaker: Dr. Leonard Creutzburg	Invited speaker
11	24.11.2026	From communication to behavior change	Jan Pfänder
12	01.12.2026	Guest speaker: Dr. Cornelia Staub	Invited speaker
13	08.12.2026	Guest speaker: Dr. Cornelia Eisenach	Invited speaker
14	15.12.2026	Discussion and lessons learnt	Viktoria Cologne & Jan Pfänder

Our guest speakers

Federal Office



Dr. Cornelia Schwier –
Head of Climate Monitoring
and Information,
MeteoSwiss

NGO



Dr. Leonard Creutzburg –
Co-lead One Planet Lab,
WWF

Industry



Dr. Cornelia Staub –
Head of Sustainability,
Migros Industrie

Journalism



Dr. Cornelia Eisenach –
Climate journalist,
Republik

Research



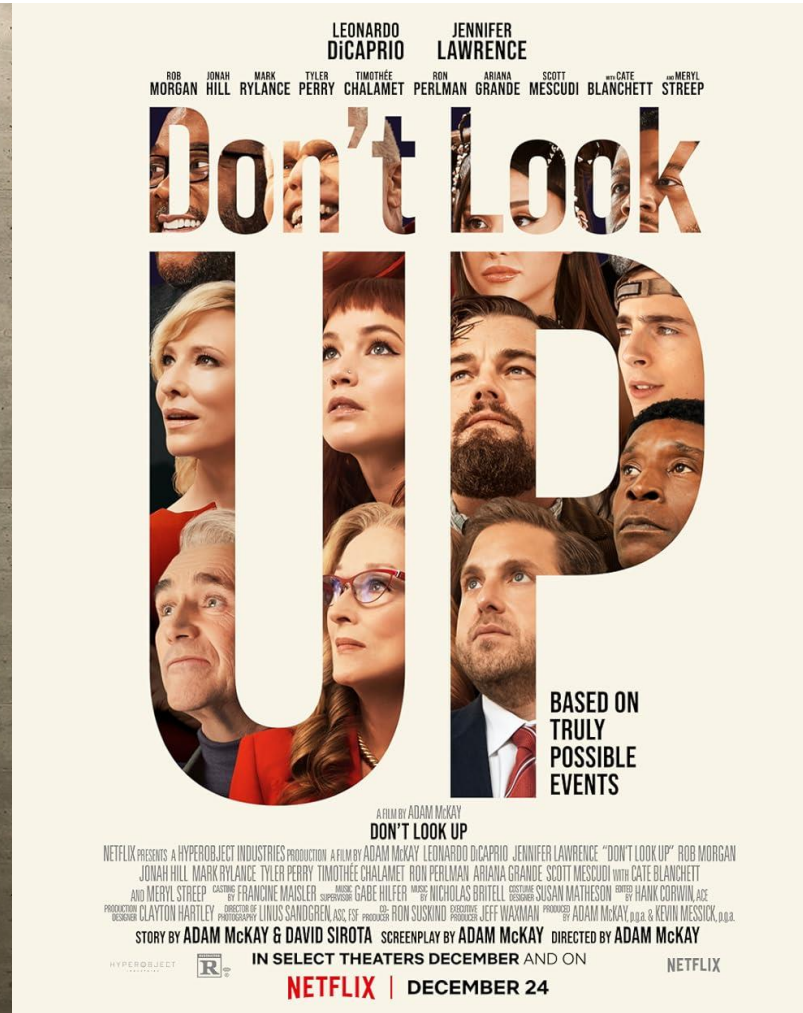
Prof. Reto Knutti –
Climate physicist,
ETH

Learning objectives

At the completion of the course, you will be able to:

- Independently and accurately identify challenges in climate change communication
- Clearly explain the psychological mechanisms that influence climate change perceptions and attitudes
- Critically apply insights from psychology to evaluate the effectiveness of climate change communication
- Develop evidence-informed recommendations for improving climate change messages for specific audiences and communication contexts.

Real-life learning objectives



Participation

- Actively participate in the course by drawing on your own reading of the materials, as well as your personal experiences and perspectives.
- Participating: asking deep and challenging questions but also asking simple clarification questions like “I’m just not getting this, please explain it in some new way” or “I’m lost, can you remind me why we’re talking about this?”.
- You are expected to read the main article assigned for each lecture.
- *When we have a guest speaker, come to class with three questions.*

Evaluation

- Not graded: pass or fail.
 - No written exam
 - Two assessments to pass the class:
- 1. Oral group presentation of an example of climate change communication**
 - 2. Written one-page reflection on the presentation by December 18**

Evaluation 1: Group presentations

- Presentations start in Week 5
- About 5 students per group
 - Exact number will depend on final enrollment
 - **Important: Participation in this course requires a course-internal commitment at latest by September 25 (23h59)**
- We will randomly assign you to groups in Moodle

Evaluation 1: Group presentations

- Structured guidelines for student presentations on Moodle
- Presentations last 10 minutes with 5 minutes of Q&A
- All group members need to actively present (approximately equal time split)
- Power point (or similar) required
- Any type of content/message that focuses on climate change can be presented as an example, such as an advertisement, political campaign, social media post (video or text), newspaper article, etc. You can present good or bad examples of climate change communication, including messages on climate change denial.
 - Be as *creative* as you want!
- You can use AI for the preparation of the presentation, but we expect you to select the examples yourself
 - ...And reflect on the use of AI in the Reflection Task

Evaluation 1: Group presentations

Presentations need to introduce the content/message and address the following questions and explicitly draw on relevant concepts learned in the course:

- Introduce your example
 - How did you come across it and why did you pick it?
- Who is the communicator and what is their motivation?
- Who is the target audience?
- What makes this a good/bad example of climate change communication?
 - Is the communicator reaching the right target audience with their message/content?
 - Is the chosen medium appropriate to support the message, and why?
- How could the message/content be improved?

Evaluation 2: Reflection task

- We want you to take some time to reflect on the group presentation.
 - Then write down your thoughts in 400-600 words and submit the reflection on Moodle **by December 18.**
 - Each group member writes their own reflection

Evaluation 2: Reflection task

- Describe how your group selected the communication example
- Clearly state your role in the project and your overall contribution to the group project
- Describe the group dynamics (e.g., Where there any conflicts? How did you manage the organization, prepare for the presentation, and decide on whether to use AI?)
- Is there anything you would do differently in hindsight?
- Reflect on your own contribution: Are you happy with your contribution/performance? Would you have liked to do more/less?
- Declare the use of AI in your group
 - If yes: How did you use it? Which model did you use? In which ways do you think using AI helped you? Do you think there are any downsides to the use of AI for your project?

Any questions?

Working with Mentimeter



What do you hope to learn from this course?



menti.com

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Waiting for participants

What is climate change communication?

“*Climate change communication* refers to any effort—explicit or otherwise—that aims to raise public awareness, understanding, and/or active engagement with the issue.” (Moser, 2010)

Different types of climate change communication



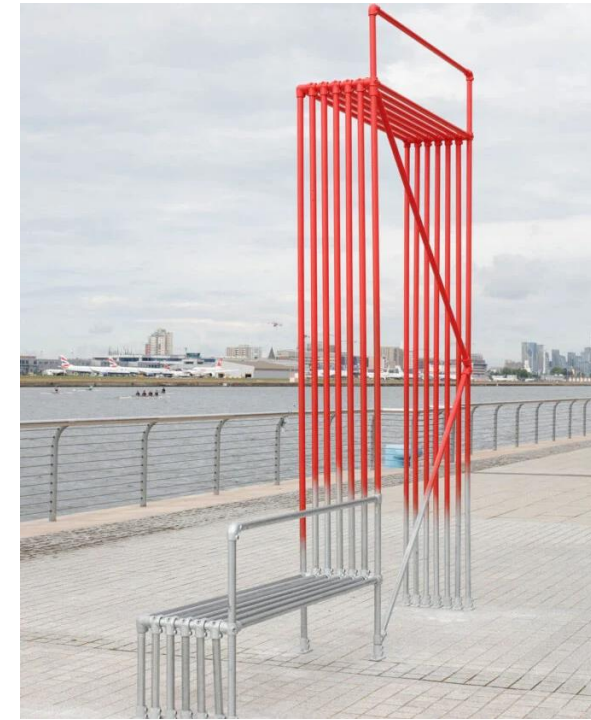
News media reporting



Campaigns



**Personal communication
with friends and family**



Art installations

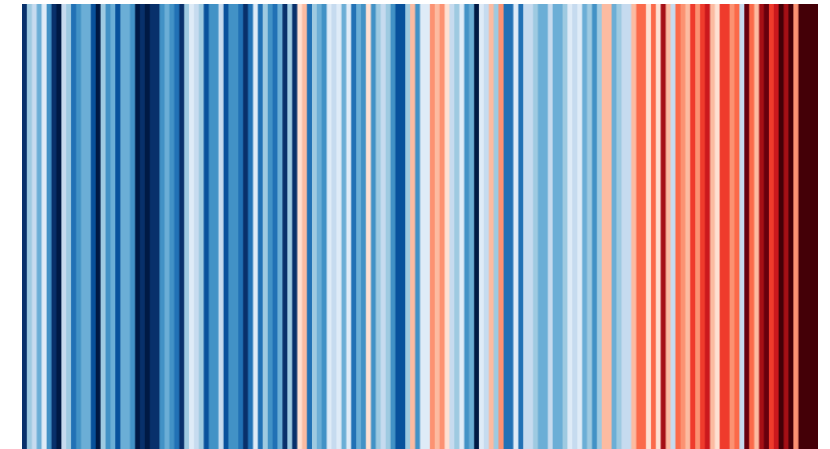
Different types of climate change communication



Videos



Social media posts



Visualizations

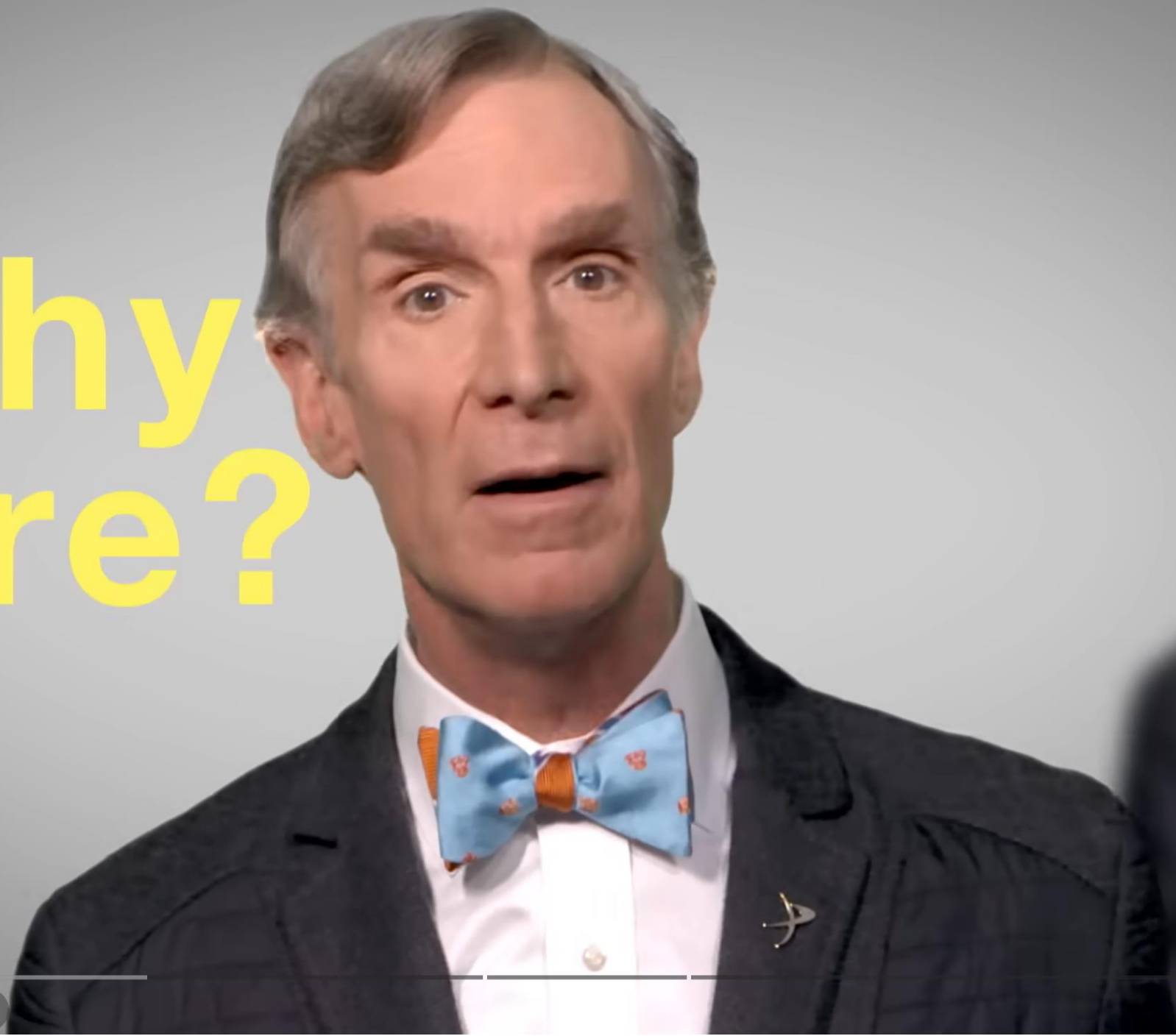
Why is climate change communication important?

Why climate change communication is important

Climate change is one of the most pressing societal issues



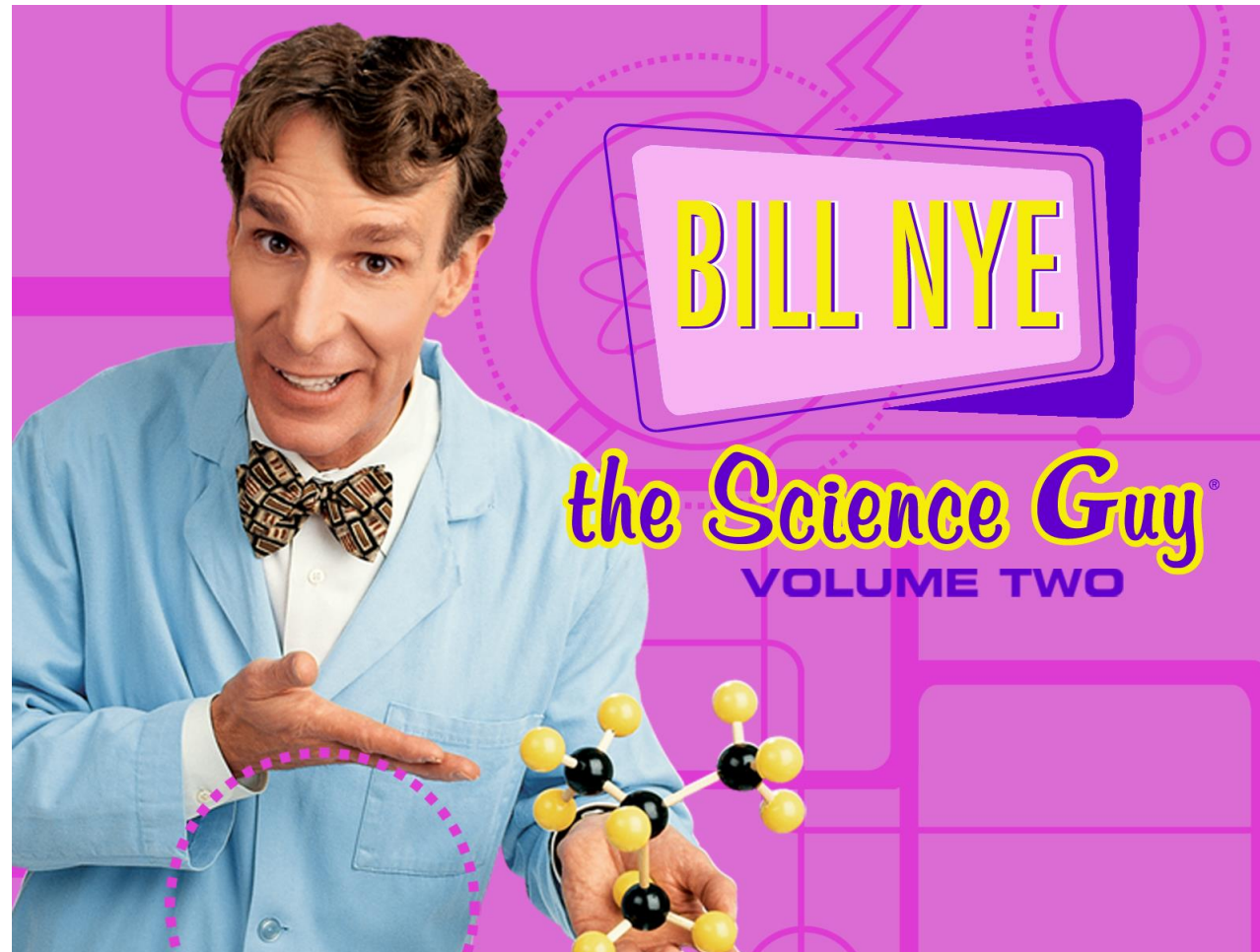
Why care?



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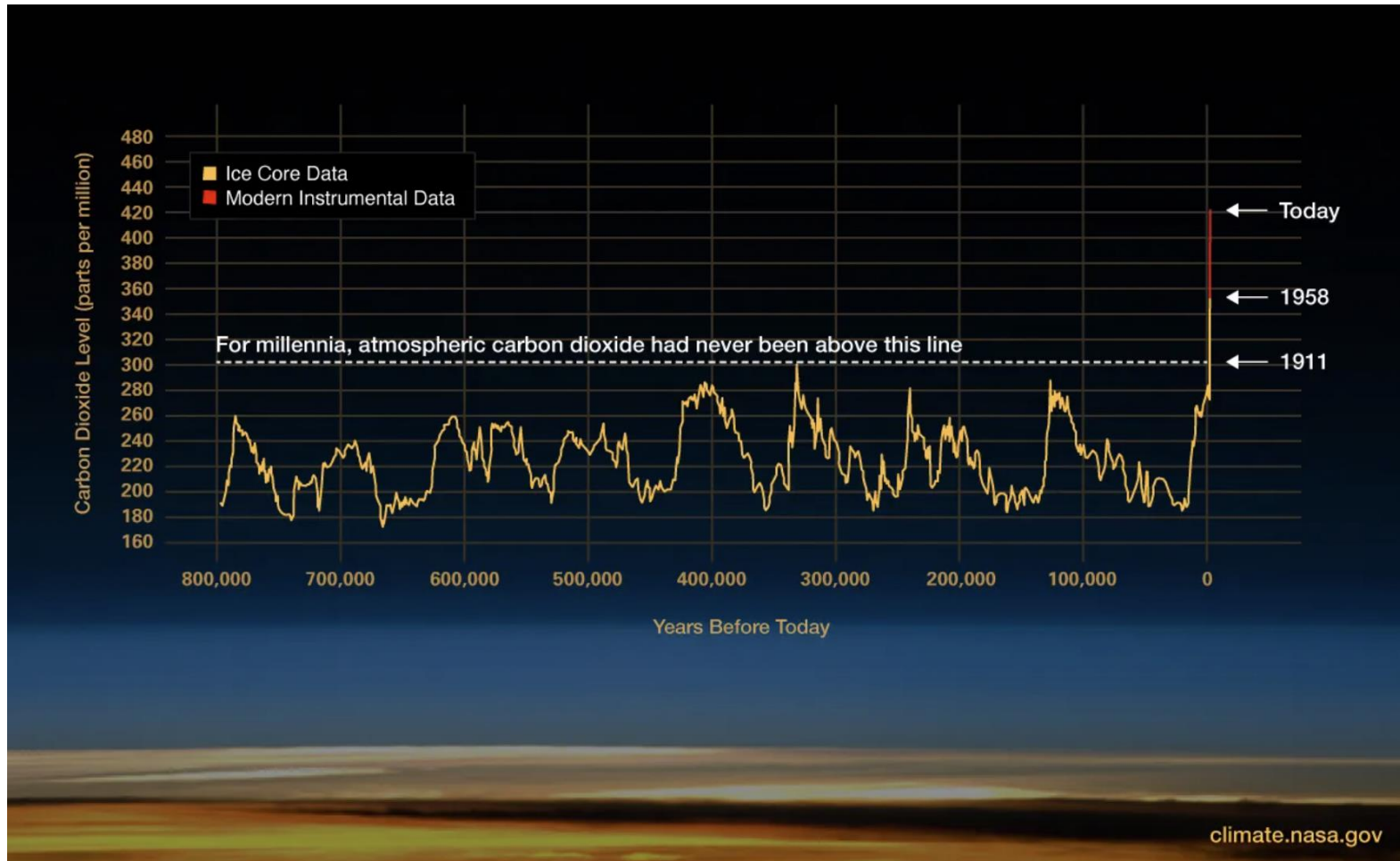
What is climate change >





Why climate change communication is important

Climate change is one of the most pressing societal issues



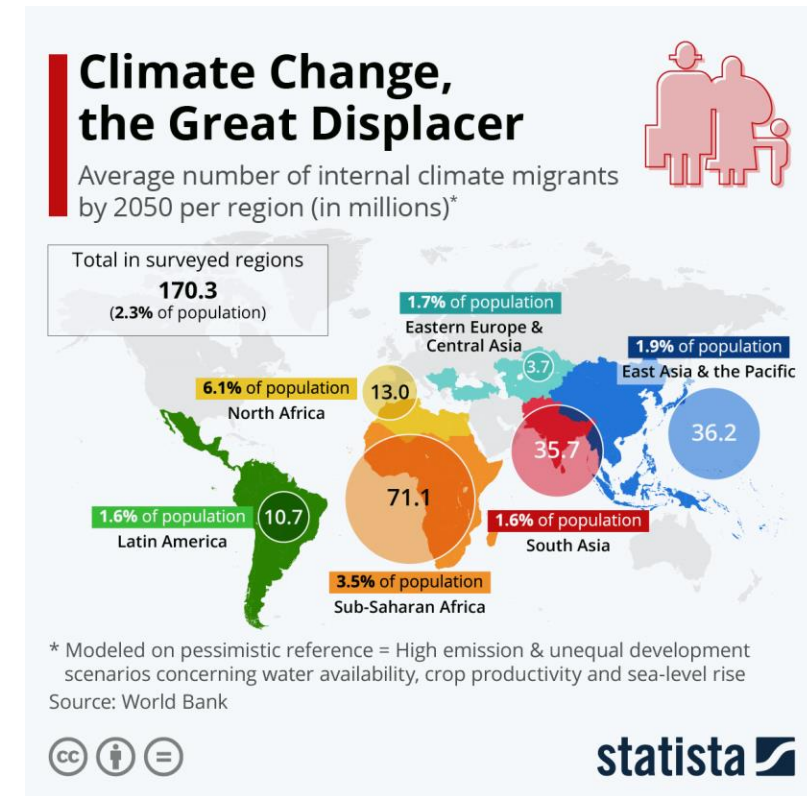
Climate change is one of the most pressing societal issues

- Extreme weather events (e.g., heatwaves, floods)
- Species extinction
- Agricultural productivity
- Sea-level rise
- Water scarcity
- Climate migration

Why climate change communication is important

Climate change is one of the most pressing societal issues

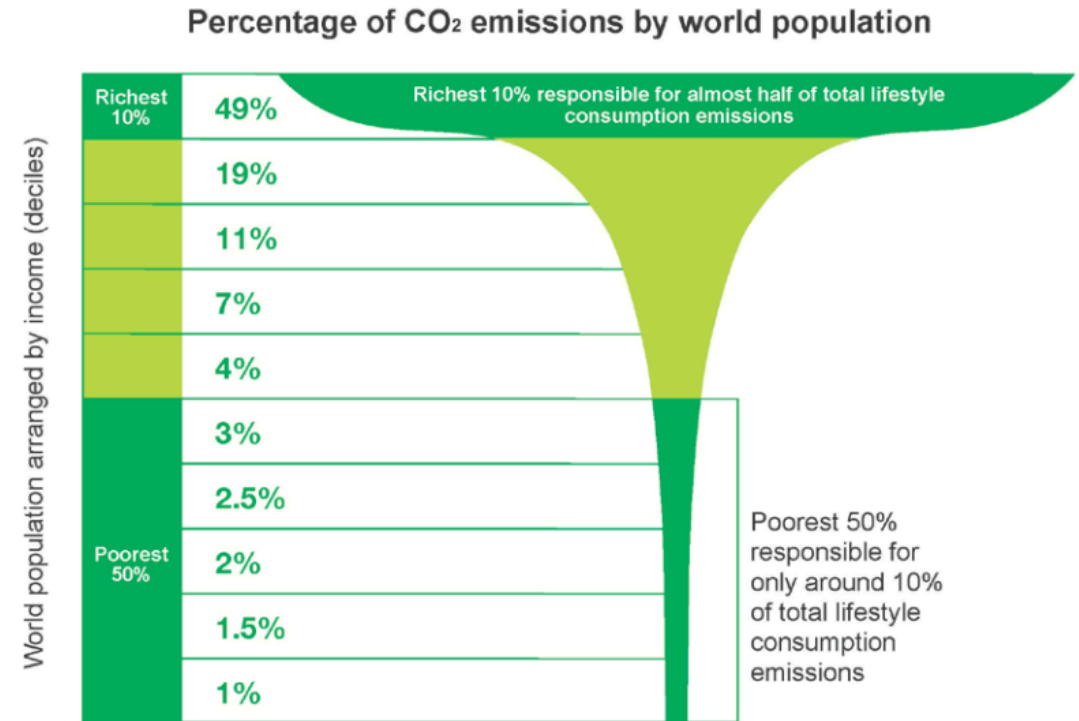
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Climate change is one of the most pressing societal issues

- Extreme weather events (e.g., heatwaves, floods)
- Species extinction
- Agricultural productivity
- Sea-level rise
- Water scarcity
- Climate migration
- Climate justice

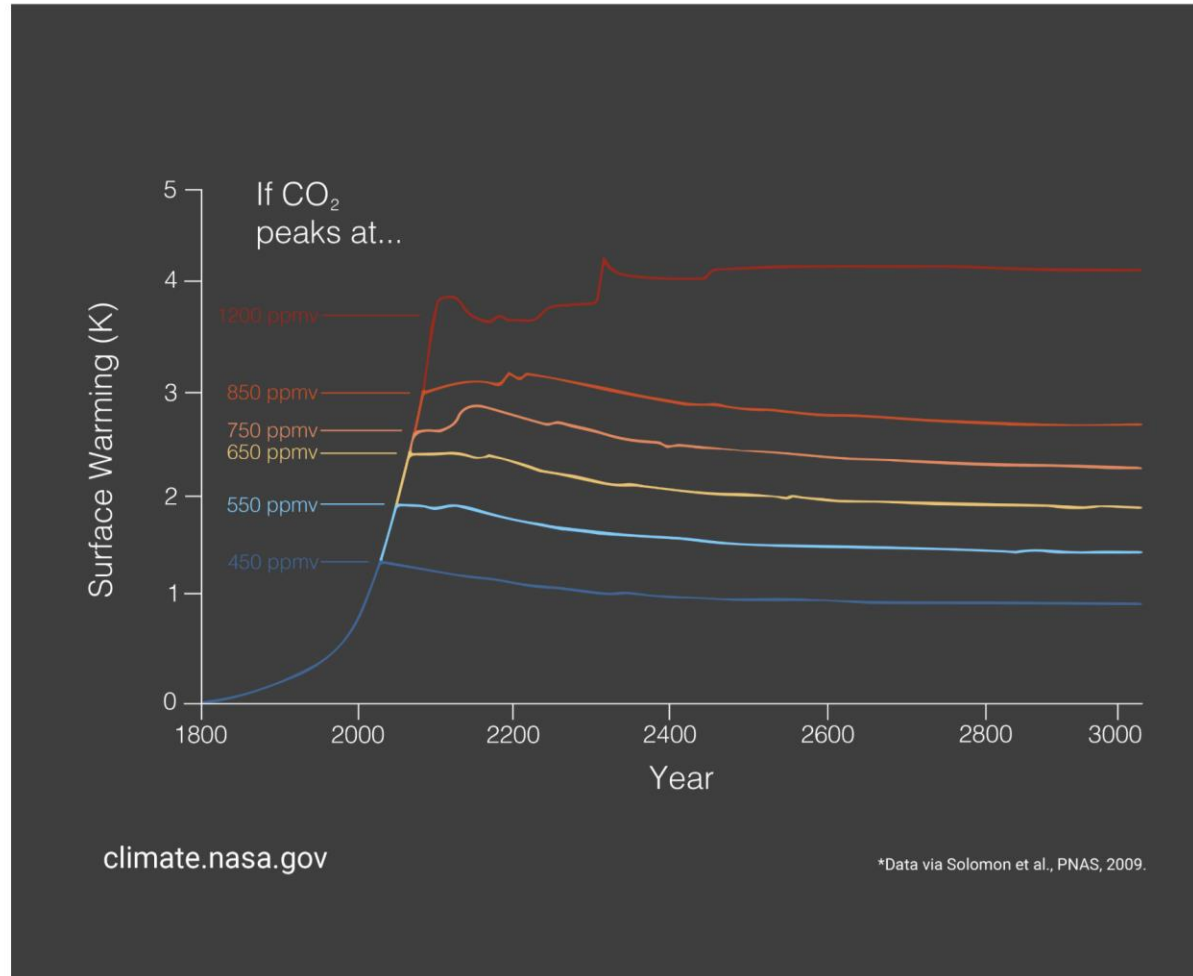
Figure 1: Global income deciles and associated lifestyle consumption emissions



Source: Oxfam

Why climate change communication is important

Good news... When emissions stop, warming stops



Why climate change communication is important

Climate change communication influences how individuals, organizations, and societies perceive climate change, and how they react to it.



Let's discuss

1. Turn to the person on your right.
2. For 2-3 minutes, discuss the following questions:
 - Why is climate change communication important?
 - Any reasons I haven't mentioned so far?

Why is climate change communication so challenging?

The problem itself

It's not an “in-your-face” type of problem (Gardiner, 2011):

- its **impacts are diffuse** and largely expected to occur in the future
- relatively few people are **directly impacted** by its effects currently (note: 2011 vs 2026)
- **attributing specific negative outcomes** to climate change is challenging
- it is largely a **side-effect** of normal, daily behavior, not intentionally caused
- it **lacks obvious culprits and personally known victims** (Markowitz & Shariff, 2012)



Psychological barriers to engagement and communication

Psychological processes shape:

- The development of attitudes and risk perceptions
- How people search for and process information
- How people ultimately make decisions

Psychological distance (Trope & Liberman, 2010)

People perceive climate change as:

- affecting other people (social distance)
 - happening in other places (spatial distance)
 - happening in the future (temporal distance)
 - uncertain (hypothetical distance)
- Climate change is deeply unrelatable at the personal level for most individuals (Spence, Poortinga, & Pidgeon, 2012). This perceived distances reduces urgency and motivation to act.

Psychological distance

Why?

- Psychological distance is egocentric: Its reference point is the self in the here and now.
- Distant things are processed abstractly rather than concretely. Abstract threats generate less emotional response and less motivation to act than concrete ones.

Psychological distance – overestimated?

Review

The psychological distance of climate change is overestimated

Anne M. van Valkengoed,^{1,*} Linda Steg,¹ and Goda Perlaviciute¹

¹Faculty of Behavioural and Social Sciences, University of Groningen, Grote Kruisstraat 2/1, 9712TS Groningen, the Netherlands

*Correspondence: a.m.van.valkengoed@rug.nl

<https://doi.org/10.1016/j.oneear.2023.03.006>

Key findings

20 recent opinion polls showed **more than 50%** of respondents believed that:

The consequences of climate change are already occurring, or will occur within the next few years.

Climate change poses risks to or is already affecting their local area or community.



Temporal distance

- Most people know that climate change is happening now.
- But they also know that the impacts are diffuse and largely expected to occur in the future.
- The consequences of our emissions and our lifestyles choices are delayed in time.
- Environmental issues present an intergenerational dilemma as they entail a tradeoff between one's own self-interest in the present and the interests of other people in the future (Wade-Benzoni & Tost, 2009).
- Cognitive mechanisms influence our perceptions of risks such as climate change.

➤ Temporal discounting

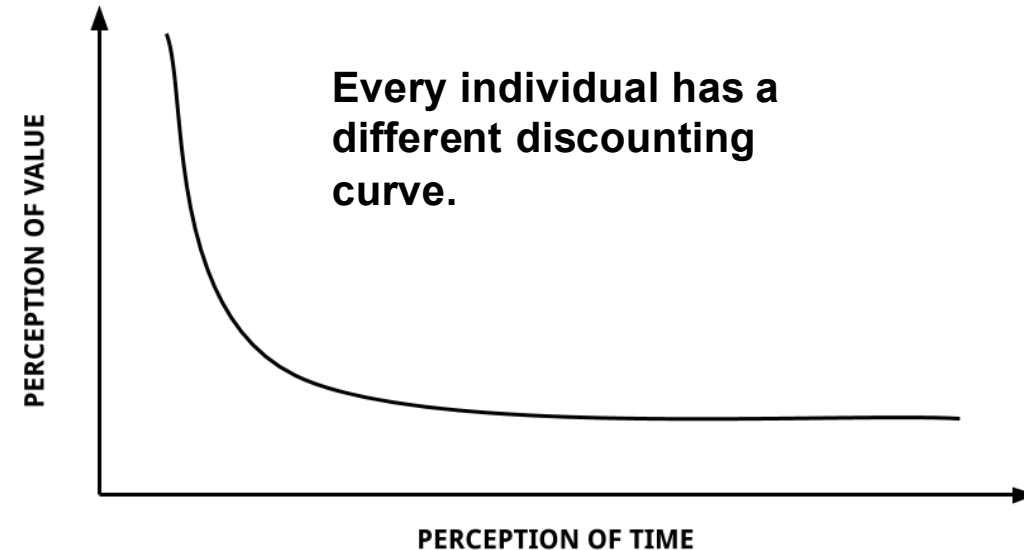
Temporal discounting

- **Temporal discounting is the general tendency to place less value on the future than on the present.**
- Given a choice between a smaller reward now and a larger reward later, people often take the smaller reward now, even when the later option is objectively better.

Why climate change doesn't feel urgent

Temporal discounting

- The discounting rate is inconsistent: we discount today vs. tomorrow much more harshly than 10 years vs. 11 years from now (hyperbolic).
 - Delay lowers the attractiveness of positive events and decreases the unpleasantness of negative ones (Laurie Hendrickx & Nicolaij, 2004)
- **The near-term sacrifice feels disproportionately painful.**



Why climate change doesn't feel urgent

Temporal discounting

- The Marshmallow test (Walter Mischel, Stanford, 1960s)



Temporal discounting

Why does it matter for climate change communication?

- Climate change: pay costs today, get benefits years or decades later.
- As we tend to focus more on the present, far-away risks seem less urgent than issues we face today.
- But: People differ in their time horizons: they can be more or less past-oriented, present-oriented, or future-oriented.
- Future-orientation is associated with climate change mitigation behavior (Milfont et al., 2012)
- Cultural differences also matter: Countries with languages that grammatically highlight the future are associated with higher climate concern (Zhu et al., 2024)

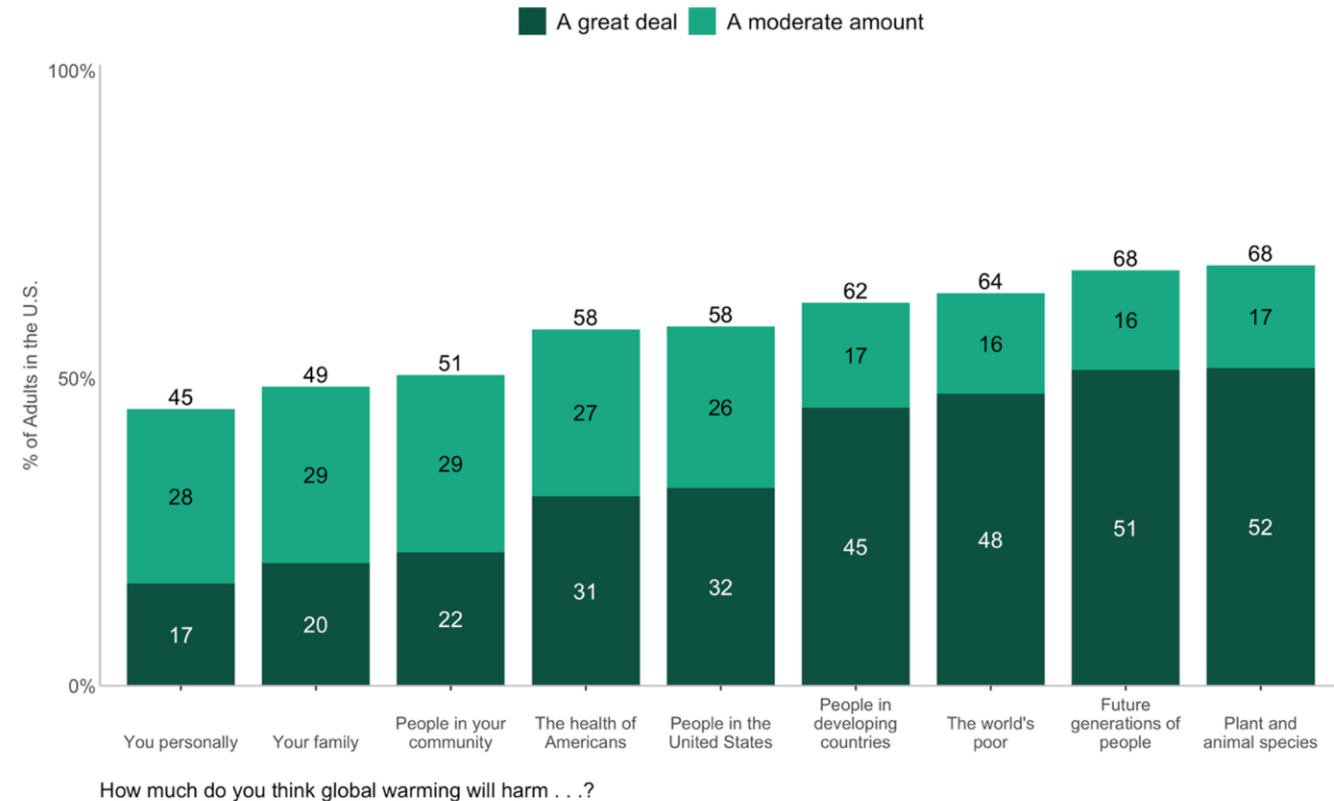
Why climate change doesn't feel urgent

Spatial distance

- People care less about risks that are perceived as distant.
- People tend to underestimate the probability that they will be affected personally by the consequences of climate change.

➤ Optimism bias

Many Americans think global warming will harm them, but more think others will be harmed



Adults in the U.S., Spring 2026

Source: Yale Program on Climate Change Communication; George Mason University Center for Climate Change Communication

Availability heuristic (Tversky & Kahneman, 1974)

People are more likely to overestimate the occurrence of an event the easier it is for them to remember examples of similar events.

- Tornadoes are seen as more frequent killers than asthma although the latter causes 20 times more deaths.
- Death by lightning is judged less likely than death from botulism even though it is 52 times more frequent.
- Climate change risks are less salient than other risks that occur more frequently and where there is more consistent media coverage.

Cognitive dissonance

- Cognitive dissonance can be understood as a “mental conflict”.
- In the presence of conflicting beliefs and behaviors, people experience **emotional discomfort** (Festinger, 1957).
- Cognitive dissonance causes a feeling of tension: a person experiences unpleasant emotions (e.g., anxiety, shame, guilt) and will strive to get rid of discomfort in two main ways:
 1. Change the belief
 2. Change the behavior

Why climate change communication is challenging

Examples of cognitive dissonance



Why climate change communication is challenging

Examples of cognitive dissonance

1. Change the behavior



2. Change beliefs



Why climate change communication is challenging

Examples of cognitive dissonance



Why climate change communication is challenging

Examples of cognitive dissonance

1. Change the behavior



2. Change beliefs



The Irish edition of the Daily Mirror, 26 June 2019

Motivated reasoning and confirmation bias

- **Motivated reasoning:** **biased information processing** in accordance with prevailing motivations and worldviews. (Kunda, 1987)
 - People access, construct, and evaluate arguments in a biased fashion to arrive at a preferred conclusion.
 - A trigger of motivated reasoning is the confrontation with a certain threat to the self.
- **Confirmation bias:** People seek out information that reinforces their **preexisting beliefs** and interpret new information to conform to preexisting beliefs.
- Information consistent with a **favored conclusion** is held to a **lower standard of scrutiny** than information consistent with an unwanted one.

Why climate change communication is challenging

Motivated reasoning and confirmation bias



Motivated reasoning and confirmation bias

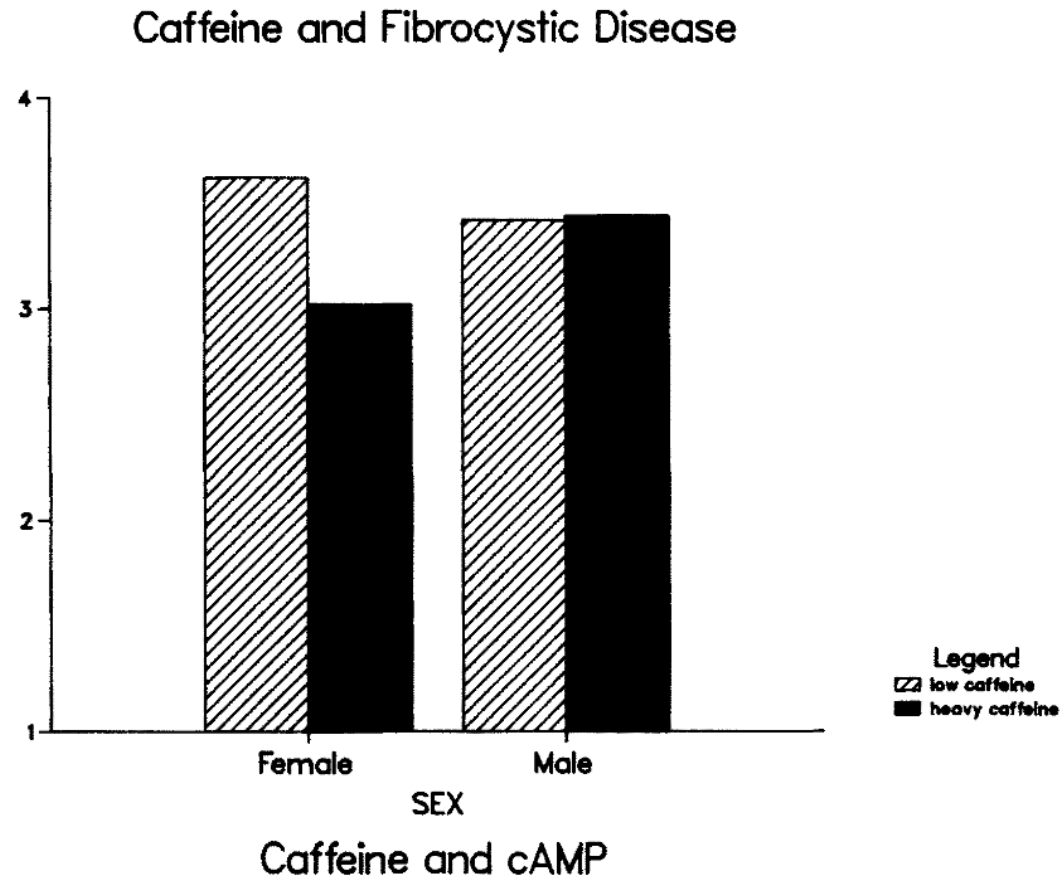


Figure 3. Study 3: Mean responses given by female and male heavy and low caffeine consumers to questions about the extent to which they were convinced by information about the negative effects of caffeine.

Understanding psychological barriers can help us improve communication about climate change and address the challenges of climate change communication.



Viktoria Cologne
@colognav

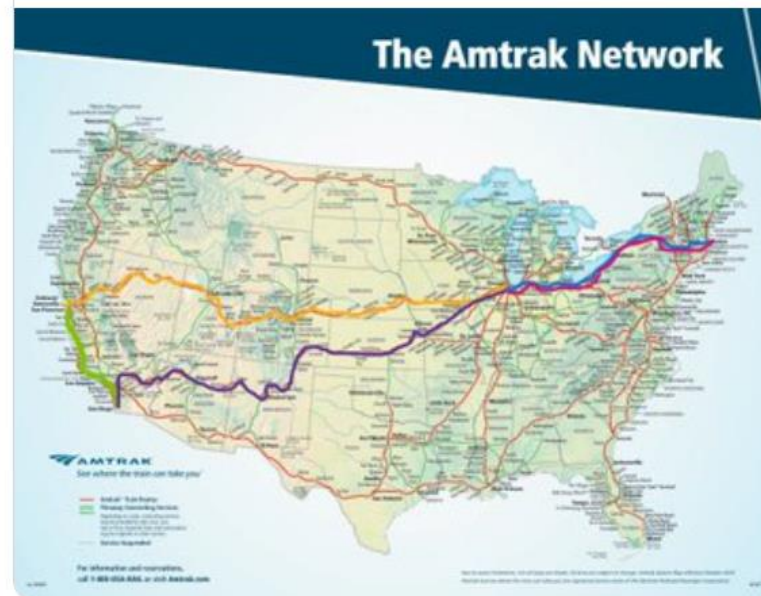
...

In Amtrak dining cars, you get randomly seated next to other people. This morning, I had breakfast with two conservatives (father and daughter from Florida) and we started talking about climate change... A thread on [#SciComm](#) with skeptical audiences 📖



Viktoria Cologne @colognav · 28 Sep 2023

Ready for our USA cross-country train trip 🚂🇺🇸
1 #USARailPass
5 stops 🏠🏠
164 hours on #Amtrak
10,800 km 🌍...
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